ELARA: Reliability-Aware Multimodal Anomaly Fusion Under Degraded Evidence

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Abstract

This chapter presents the final audited evidence for ELARA, a reliability-aware multimodal score-fusion architecture with Reliability-Gated Attention (RGA), evaluated under controlled degradation and strict claim governance. In the powered static-reference study (Family A), validation-frozen RGA+ improves over fixed static attention in all five previously inspected cells (A-POWERED-1: ΔAUC $+0.1082$, Holm $p = 3.35 \times 10^{-4}$; A-POWERED-2: $+0.0519$, Holm $p = 4.06 \times 10^{-5}$; A-POWERED-3: $+0.1038$, Holm $p = 4.06 \times 10^{-5}$; A-POWERED-4: $+0.0297$, Holm $p = 1.53 \times 10^{-3}$; A-POWERED-5: $+0.0095$, Holm $p < 10^{-15}$, practically small versus the Phase-1 estimate of $+0.0003$). Cross-category transfer remains near chance, and LOCO-AD and VisA conclusions remain bounded by derived-view-proxy pairing.

In mechanism replication (Family B), RGA reproduces the zero-attack coherent score-collapse endpoint (B1: $+0.0507$ versus Phase-1 target $+0.0506$, 95% CI [$+0.0364$, $+0.0650$], Holm $p = 4.31 \times 10^{-12}$) and retains a positive maximum-attack result (B2: $+0.0939$ versus $+0.0319$, 95% CI [$+0.0741$, $+0.1149$], Holm $p < 10^{-15}$) with explicit estimator-change qualification. Sensitive RGA-v2 variants fail clean false-fire controls (firing rate 1.000), and domain composition shift audits (B-MECH-3S) do not reduce false-fire rates. Monitoring and retrospective certification outcomes are mixed.

The held-out Eyecandies transfer study (Family D) satisfies the clean false-fire budget (0.000 observed under validation-selected per-cell thresholds $\tau = 0.52$ and $\tau = 0.51$) but is not confirmed under primary bootstrap interval inference: the primary paired-bootstrap ensemble Δ AUC is -0.0010 (95% CI [-0.0114, $+0.0092$]) for D-EYE-1 and -0.0109 (95% CI [-0.0254, $+0.0034$]) for D-EYE-2, both intervals spanning zero, and a secondary per-seed paired t-test is non-significant ($p = 0.3632$ and $p = 0.4468$). A new D13 natural clean-transfer track adds development evidence for a stronger target: a validation-only residual-stack candidate beats both SAR ($\Delta = +0.0583$, 95% CI $[+0.0410, +0.0759]$) and confidence-weighted mean ($\Delta = +0.0182$, 95% CI $[+0.0075, +0.0290]$) on opened 3D-ADAM, while MulSen remains unresolved. Because both datasets are opened, D13 is integrated as pending fresh-holdout evidence and not as final Gate E readiness. The chapter therefore supports a bounded mechanism claim and identifies calibration transfer under score-distribution shift as the main unresolved limitation.

1 Introduction and Motivation

Operational anomaly detection is rarely a single-model problem. A fraud case may appear in transaction features, device behavior, account history, text evidence, or network activity. An industrial defect may appear in both color information and geometry. A security event may be weak in one log source and clearer in another. In these settings, the system designer must decide not only which domain models to train, but also how to combine their outputs once those models are deployed.

This chapter focuses on the second problem: score-level fusion under imperfect domain reliability. The assumption is that domain experts already produce scores, confidence values, and compact evidence vectors. The open question is how the fusion layer should behave when domain quality changes at inference time. A static fusion model can learn useful interactions from training data, but it may continue to trust a domain after that domain has shifted, collapsed, or become poorly calibrated. A simple confidence-weighted average is easy to understand, but it uses local confidence values without asking whether the domain itself still resembles the validation distribution.

ELARA was built to study this problem in a reproducible way. It is not presented as a new fraud detector, cybersecurity detector, text detector, or industrial inspection detector. Instead, it is a research system for combining heterogeneous anomaly evidence. The central component is RGA, a reliability-aware fusion module that uses three forms of evidence: validation calibration, test-time score-distribution drift, and score sharpness. The module is deliberately conservative. When domains appear reliable, the system uses the static attention model. When the average reliability drops below a threshold, the reliability-aware path is activated.

The chapter is in this form because the current evidence is both useful and bounded. The real-domain fusion benchmark combines fraud, cyber, behavior, and text datasets, but those records are aligned by label rather than by shared entities, timestamps, or incidents. It is useful for score-fusion stress testing, but it is not a complete multimodal incident benchmark. To reduce this gap, the repository now includes an all-category MVTec 3D-AD RGB/3D benchmark path. That path gives natural co-observation, while its canonical one-class protocol and lightweight score construction make it a protocol diagnostic rather than a final performance claim.

1.1 Research Question

The research question is:

> *Can a multimodal anomaly-fusion system preserve competitive clean score-fusion performance while using calibration and drift evidence to reduce brittle behavior under domain reliability stress?*

This question is intentionally narrower than broad anomaly-detection superiority. The chapter does not claim that RGA is always better than static attention, that the current paired MVTec run is complete, or that the synthetically paired benchmark represents naturally co-observed incidents. The claim is about the fusion layer under specific stress conditions.

1.2 Main Claim

The main thesis of this chapter is:

> *RGA is most useful as a stress-response mechanism: it preserves the static attention path when reliability evidence is high, and it helps under coherent score-collapse conditions when reliability evidence indicates domain failure.*

The measured evidence supports this claim on the secondary benchmark. The clean table is near saturated and should be treated as a sanity check. The reliability-stress experiments, the threshold sweep, and the component ablation provide the more useful evidence about the mechanism.

1.3 Contributions

We make five scoped contributions:

1. We present ELARA, a score-level anomaly-fusion research system with a unified long-format schema for heterogeneous evidence.
2. We define RGA, a conservative reliability-gated attention module that combines masked attention with calibration, drift, and sharpness signals.
3. We evaluate the method against practical score-fusion baselines, including random forest fusion, early-fusion MLP, late-fusion ensemble, confidence-weighted mean, score-level Tent, and pseudo-label TTT.
4. We report an all-category paired MVTec 3D-AD benchmark, multi-seed confidence intervals, mechanism ablations, failure cases, and explicit threats to validity.
5. We add the D13 natural positive-transfer track, which requires a validation-only candidate to beat both SAR and confidence-weighted mean on a fresh natural holdout; opened 3D-ADAM and MulSen results are retained as development evidence only.

1.4 Evidence Boundary

The evidence boundary is part of the contribution because it prevents the results from being overstated. The real-domain benchmark uses real local datasets, but the records are label-aligned. The MVTec benchmark uses natural RGB/3D pairing across eight categories, but it still uses lightweight normal-reference image statistics and the canonical one-class anomaly protocol. These two artifacts answer different questions. The first is useful for reliability-stress score fusion. The second is useful for exposing protocol limits on naturally paired data. Neither should be treated as a complete final benchmark for all multimodal anomaly detection.

2 Background and Related Work

2.1 Anomaly Detection as a Systems Problem

Deep anomaly detection has been studied across tabular, sequential, image, text, and graph data. A common difficulty is that the definition of an anomaly depends on the application, the label source, the imbalance level, and the cost of false positives and false negatives. Pang et al. [3] emphasize that anomaly detection methods cannot be evaluated without attention to context. This chapter follows that view. The purpose is not to replace domain-specific detectors. The purpose is to study what happens after those detectors produce evidence that must be fused.

2.2 Multimodal and Score-Level Fusion

Multimodal learning is often described in terms of early fusion, late fusion, hybrid fusion, and interaction-based fusion [2]. Early fusion joins features before prediction. Late fusion combines decisions or scores from separate models. Stacking trains a meta-model over domain outputs. Attention-based fusion gives the model a way to learn interactions among domains, especially when some domains are missing. The weakness is that standard attention learns a trust policy from the training distribution and does not automatically know when a domain has become unreliable at inference time.

The score-level setting used in this chapter is intentionally practical. Many deployed systems cannot easily retrain all domain experts together. Different teams may own different detectors, and those detectors may expose only scores, calibrated probabilities, or compact embeddings. A score-level fusion study is therefore narrower than raw multimodal learning, but it matches a common engineering constraint.

2.3 Calibration, Drift, and Reliability

Calibration asks whether predicted probabilities match empirical frequencies. Post-hoc calibration methods such as Platt scaling, isotonic calibration, and modern neural calibration analyses show that probability quality is separate from ranking quality [4, 5, 6]. Uncertainty and calibration can degrade under dataset shift [7]. For a fusion system, this matters because a domain can remain numerically confident while becoming less trustworthy.

RGA treats calibration as one signal, not as the whole reliability definition. The reliability estimator also compares current score distributions against validation score distributions using a Kolmogorov-Smirnov statistic and includes a score-sharpness term. This design is not meant to be a complete theory of reliability. It is a simple, measurable rule that can be tested under controlled stress.

2.4 Test-Time Adaptation Baselines

Test-time adaptation updates a model during inference to handle distribution shift without access to labels at test time. Two threads of this literature matter for this work, and both are represented in the baseline suite.

Test-time training (TTT). Sun et al. [8] proposed updating a model on each test sample by minimizing a self-supervised auxiliary loss (the original work used rotation prediction). The intuition: an auxiliary task trained jointly with the main loss anchors the representation, so adapting the auxiliary component at test time also adjusts the main predictor in a useful direction. Later variants relax the self-supervision to pseudo-labels: the model produces a confident prediction on the test sample, treats it as the label, and updates the parameters with a small step against that target. The pseudo-label adapter in this study (`ttt_pseudo_label_adapter`) is a score-level version of this approach: it acts on the fusion-head logits rather than on a deep representation.

Entropy-minimization TTA. Wang et al. [9] introduced Tent, which adapts only the batch normalization affine parameters of a frozen model to minimize the entropy of test predictions. Tent is simpler than full TTT and is the cleanest articulation of the hypothesis that confident predictions are correct predictions — formally, that the local entropy of the predictive distribution is a sufficient signal for which direction to update. The Tent score adapter in this study (`tent_score_adapter`) implements the score-level analog: a frozen fusion head is adapted by a scalar temperature and bias fitted to minimize per-sample entropy on the test batch.

EATA and SAR. The score-level baseline suite also includes EATA [22], which adds reliable-sample selection and non-redundancy filtering on top of Tent, and SAR [23], which adds per-sample entropy weighting with sharpness-aware minimisation on top of Tent. Both are implemented as score-level adapters operating on the static fusion head's logits; they appear in every result table that lists the test-time-adaptation comparator family.

Position of RGA relative to TTA. Both Tent and TTT adapt the *prediction* at test time — they ask which output should be revised. RGA sits one level up: it adapts the *weighting* of upstream domain scorers, not the fusion head itself. The reliability signals (validation ECE, KS drift, sharpness) are properties of the upstream domains' behavior, and the gate decides whether to trust them collectively or to fall back to a static weighting learned from training data. This makes RGA conceptually closer to selective prediction [17] — a controller that decides when to commit to the model's standard behavior — than to TTA in the Tent sense. Empirically the chapter compares all three (Tent, TTT, RGA) on both benchmarks so the contrast is visible.

2.5 RGB–3D Industrial Anomaly Detection

MVTec 3D-AD provides naturally paired RGB and 3D observations for industrial anomaly detection and localization [11]. Recent work on RGB–3D anomaly detection has shown that paired color and geometry can be useful when the detector is designed for that modality pair. M3DM uses hybrid feature and decision fusion with memory banks [12]. Crossmodal feature mapping uses agreement between RGB and point-cloud features as an anomaly cue [13]. Real3D-AD expands point-cloud anomaly benchmarking [14], and M3DM-NR studies noisy-resistant multimodal anomaly detection [15].

The MVTec path in this repository is not a replacement for those specialized RGB–3D detectors. It is a paired-data input path for the score-fusion layer. This distinction is important. The current MVTec smoke run asks whether the repository can construct and evaluate paired score-fusion inputs. It does not claim that the lightweight scorers compete with specialized RGB–3D methods.

3 System and Method

3.1 System Boundary

ELARA is organized around the idea that each sample may have several domain observations. A domain can be fraud, cyber telemetry, behavior, text, RGB imagery, depth/XYZ geometry, or another source that can produce a score and compact evidence. The fusion layer receives these domain rows in a common schema. It then produces one anomaly probability and optional domain-level attributions.

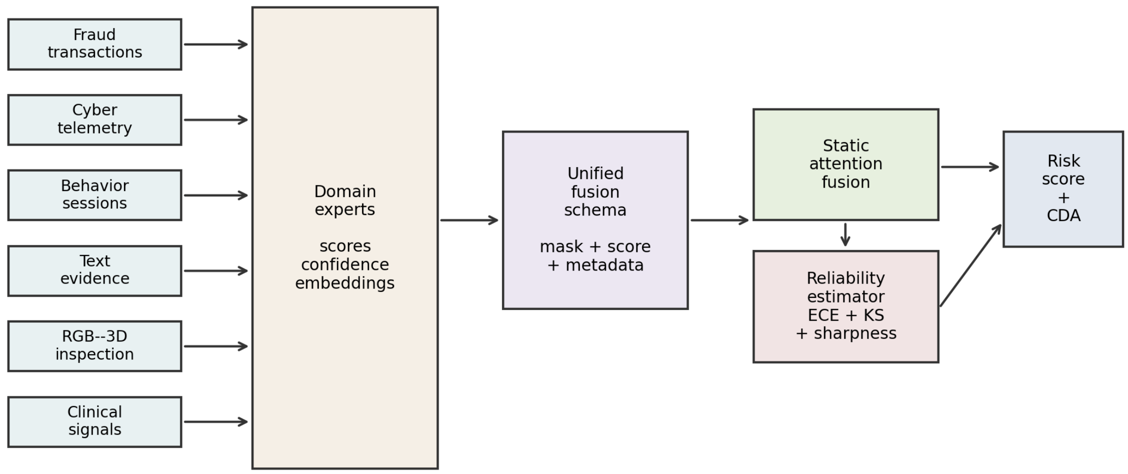

Figure 1: ELARA system architecture. Domain experts produce scores, confidence values, and compact embeddings. The fusion schema feeds static attention and the reliability estimator. The output is a risk score plus counterfactual domain attribution.

The system design separates domain learning from fusion learning. This is a deliberate simplification. It lets the research focus on reliability-aware fusion without requiring every raw domain model to be retrained as part of one large multimodal network.

3.2 Input Schema

The long-format fusion schema contains one row per sample-domain pair. Each row contains a sample identifier, a domain name, a binary label when available, a score, a confidence value, and embedding columns. Missing domains are represented by masks, not by fabricated raw observations. This allows a sample with all domains, a sample with one missing domain, and a sample with only one available domain to share the same model interface.

For the label-aligned real-domain benchmark, the metadata records the field `natural_pairing` as `false`. For the MVTec 3D-AD path, the same field is `true` because the RGB and 3D observations come from the same object scan. This metadata distinction is important because the same fusion code can be used for two different kinds of evidence.

3.3 Problem Formulation

Let sample i contain up to D domain observations. Domain d contributes a feature vector $x_{i,d} \in \mathbb{R}^F$ and a missing-domain mask $m_{i,d}$. The system estimates

$$\hat{p}_i = P(y_i = 1 \mid \{x_{i,d}, m_{i,d}\}_{d=1}^{D}), \tag{1}$$

where $\hat{p}_i$ is the final anomaly probability. A static fusion model learns this relationship from the training data and uses the same learned trust policy at inference time. RGA adds a batch-level reliability vector $r \in [0, 1]^D$ in the reported runs and applies it through each sample's observed-domain mask. A per-sample reliability mode is implemented for ablation, but the tables in this chapter use the batch-level gate.

3.4 Static Attention Path

The static attention path encodes each domain vector into a shared latent space. A learned domain embedding allows the model to distinguish evidence sources. A multi-head self-attention block [1] then operates across available domains while ignoring missing domains through the mask. This path is the default because reliability adaptation should not disturb predictions when the domains look normal.

3.5 Reliability Estimation

The reliability estimator is fit after model training. For each domain it stores validation score distributions and calibration information. At inference time, the reliability for domain d is

$$r_d = \alpha(1 - \mathrm{ECE}_d) + \beta p_{\mathrm{KS},d} + \gamma S_d, \tag{2}$$

where ECE_d is validation expected calibration error, $p_{\mathrm{KS},d}$ is the Kolmogorov-Smirnov score-drift p-value, and S_d is score sharpness based on squared distance from 0.5. The current configuration uses $(\alpha, \beta, \gamma) = (0.45, 0.35, 0.20)$.

This formula is simple by design. It is not a final answer to reliability estimation. It is a testable rule that separates three observable signals: how well a domain calibrated on validation data, whether the current batch resembles the validation score distribution, and whether scores are informative or close to the uninformative midpoint.

3.6 Reliability Gate

The gate is the main difference between RGA and always-on reliability weighting. Let $\bar{r}$ be the mean reliability over present domains and let $\tau = 0.66$. If $\bar{r} \geq \tau$, the system uses the static attention path. If $\bar{r} < \tau$, reliability weights are injected into the fusion block:

$$\tilde{r}_{i,d} = (1 - m_{i,d}) r_d. \tag{3}$$

Equivalently, if unreliability is written as $u = 1 - \bar{r}$, the same rule is $u > 1 - \tau = 0.34$. The unreliability threshold is therefore not 0.66; using the same numeric threshold for both $\bar{r}$ and u would reverse the gate logic.

This design is conservative. It assumes that static attention is already a strong clean model and that reliability weights should be used only when there is evidence of degradation. This also makes the method easier to criticize and test: if the gate rarely activates, the method should behave like static attention; if the gate activates under stress, the stress outcomes should change.

The codebase also now exposes an RGA-v2 gate family for follow-up experiments: a minimum-reliability gate $G_{\min} = \mathbb{1}[\min_d r_d < \tau_{\min}]$ and a hybrid gate $G_{\mathrm{hyb}} = \mathbb{1}[\bar{r} < \tau_{\mathrm{mean}} \text{ or } \min_d r_d < \tau_{\min}]$. These variants are diagnostic candidates in the current chapter, not validated fixes. The completed k-of-D run shows that with $\tau_{\min} = 0.34$, observed minimum reliability stays above the minimum threshold, so the minimum branch never activates and the hybrid gate behaves like the original mean gate. The reported base-RGA tables keep the original mean gate unless a configuration explicitly selects `minimum` or `hybrid`.

3.7 Switching-Rule Certificate and Scope

The deployment question is whether the gated policy is preferable to static attention on the cases where the gate actually fires. Let $\ell_s(i)$ be the loss of the static path, $\ell_r(i)$ the loss of the reliability-aware path, and $g_i \in \{0, 1\}$ the gate decision. The realized gated loss is

$$\ell_g(i) = (1 - g_i)\ell_s(i) + g_i\ell_r(i). \tag{4}$$

On an audited validation window, the switch is certified only when the fired subset has a positive lower confidence bound on paired benefit. For $X_i = \ell_s(i) - \ell_r(i)$ and $\mathcal{F} = \{i : g_i = 1\}$, the empirical benefit is

$$\hat{\mu}_{\mathcal{F}} = \frac{1}{|\mathcal{F}|}\sum_{i\in\mathcal{F}} X_i. \tag{5}$$

The switch is certified only if

$$\mathrm{LCB}_\alpha = \hat{\mu}_{\mathcal{F}} - \mathrm{margin}_\alpha > 0. \tag{6}$$

The implementation is `bounded_switching_certificate`. This is a finite-sample deployment audit, not a universal population-level proof. The population theorem in Appendix C is instead stated as a risk-dominance condition with explicit detection power, false-fire rate, benefit, and clean-operation cost terms.

3.8 Counterfactual Domain Attribution

The explanation path masks one available domain at a time and recomputes the prediction. For sample i, let $\hat{p}_i$ be the original prediction and $\hat{p}_{i,-d}$ be the prediction when domain d is masked. The attribution is

$$\Delta_{i,d} = \hat{p}_i - \hat{p}_{i,-d}. \tag{7}$$

A positive value means that the domain increased anomaly risk. A negative value means that the domain reduced anomaly risk. This is not a feature-level explanation; it is a domain-level explanation for analysts who need to know which source of evidence mattered most.

4 Benchmark Construction

4.1 Naturally Paired MVTec 3D-AD Benchmark

The paired benchmark path uses MVTec 3D-AD because each object observation has paired RGB and 3D/XYZ evidence. The preparation script discovers paired files, emits two rows per observation, stores the original pairing key, and marks the metadata as naturally paired. The current run covers all eight extracted object categories – bagel, cable_gland, cookie, dowel, foam, peach, rope, and tire – for a total of 3,226 paired samples and 6,452 domain rows, with a positive fraction of 0.224 and full RGB and depth coverage.

Table 1: Naturally paired MVTec 3D-AD multi-category metadata.

Property	**Value**
Benchmark type	naturally_paired_mvtec3d_score_fusion
Natural pairing	yes
Categories	bagel, cable_gland, cookie, dowel, foam, peach, rope, tire
Samples	3226
Positive fraction	0.224
Score fit split	train/good
Score fit samples	2070
Score normalization	train_good_distance_minmax_clipped
Domains	rgb, depth_or_xyz

The initial MVTec domain scores are lightweight normal-reference image-statistic scores fit from original train-good observations only. This keeps the benchmark locally runnable, but it limits the absolute performance ceiling. The corrected attention-fusion split follows the canonical MVTec one-class protocol: train/validation rows are normal-only and the test fold is mixed. The result in Table 2 is therefore a protocol diagnostic rather than a conventional supervised leaderboard. Random forest, late fusion, Tent, TTT, static attention, and RGA have little two-class training signal; the small RGA delta should not be read as a general win. The held-out-category and M3DM-style variants probe the same limitation under different paired-data conditions. The updated held-out-category protocol now reserves positive validation rows from train categories, which lets RGA+ tune without seeing the held-out categories; it improves ROC-AUC slightly but does not settle PR-AUC or category transfer. These variants remain part of the evidence boundary discussed in §5.5.

Table 2: Naturally paired MVTec 3D-AD clean held-out performance across eight categories. Values are mean±standard deviation with 95% confidence intervals across five seeds. Under the canonical one-class protocol, this table is a supervised-fusion protocol diagnostic rather than a two-class leaderboard.

Method	**ROC-AUC mean**	**ROC-AUC 95% CI**
Random forest	0.5000	[0.5000, 0.5000]
Conf.-weighted mean	0.4458	[0.4458, 0.4458]
Tent score adapter	0.5000	[0.5000, 0.5000]
TTT pseudo-label	0.5000	[0.5000, 0.5000]
Early fusion MLP	0.5455	[0.5180, 0.5836]
Late fusion ensemble	0.5000	[0.5000, 0.5000]
Static attention	0.5424	[0.4992, 0.5661]
RGA attention	0.5609	[0.5394, 0.5842]
RGA+ boosted fusion	0.5000	[0.5000, 0.5000]
RGA+ router	0.5609	[0.5394, 0.5842]

4.2 Secondary Label-Aligned Real-Domain Benchmark

The secondary benchmark, ELARA-Bench-LA, combines four real local domains: credit card fraud, cyber telemetry, online behavior, and labeled news text. Domain scorers are trained first, and their out-of-fold scores are converted into the common fusion schema. The dataset contains 8,000 composite samples, four domains, eight embedding features per domain, a 0.307 positive rate, and 11.8% nominal missingness. Domain coverage is approximately balanced across the four domains.

Table 3: Benchmark metadata for the secondary synthetically paired fusion input.

Property	**Value**
Benchmark type	label_aligned_real_domain_score_fusion
Natural pairing	no
Pairing unit	binary-label-aligned composite sample
Samples	8000
Positive fraction	0.301
Scorer train fraction	1.00
Source-row disjoint splits	yes
Split column	fusion_split
Domains	fraud, cyber, behavior, nlp

Table 4: Out-of-fold domain scorer quality used to construct ELARA-Bench-LA.

Domain	**Rows**	**Pos. rate**	**OOF ROC**	**OOF PR**
fraud	20000	0.025	0.9686	0.7826
cyber	20000	0.500	0.9684	0.9696
behavior	12330	0.155	0.8711	0.6042
nlp	20000	0.500	0.9973	0.9973

This benchmark is useful because it uses real domain datasets and lets the fusion layer be tested under controlled stress. It is also limited because the domains are the secondary benchmark rather than naturally co-observed. A positive composite draws positive records from available domains, and a negative composite draws negative records. This creates a meaningful score-fusion stress test, but it does not teach the model real temporal or causal relationships among the domains. Source rows are now split into train, validation, and test pools before composite sampling, and the experiment runner consumes the `fusion_split` column to keep the same `domain/source_row/label` key out of multiple supervised fusion splits.

4.3 UNSW-NB15 Naturally Structured Event-View Benchmark

UNSW-NB15 provides an additional naturally structured event-view evaluation outside visual paired data. Each row is a single network event with three software-derived view modalities:

flow timing / throughput / inter-packet statistics (13 features)

connection protocol and session structure (14 features)

context aggregated past-window connection counters (12 features)

The three views are computed from the same event identifier; their pairing strength under the Phase-0.6 classification is `naturally_structured_views` rather than `independent_modalities`. Splits are stratified random draws on (attack category, label) using event id; the test fold is disjoint from train and validation.

Table 5: UNSW-NB15 five-seed clean ROC-AUC over three software-derived views (flow / connection / context) of the same network event, 55,491 paired events. Reported descriptively; method ordering follows the clean-table convention.

Method	**ROC-AUC**	**PR-AUC**	**F1**	**ECE**	**Brier**
Random forest	0.9890±0.0001 [0.9889, 0.9891]	0.9939 [0.9939, 0.9940]	0.9522±0.0002 [0.9520, 0.9524]	0.0172±0.0003 [0.0169, 0.0176]	0.0420±0.0001 [0.0419, 0.0422]
Conf.-weighted mean	0.3486 [0.3486, 0.3486]	0.5454 [0.5454, 0.5454]	0.7798 [0.7798, 0.7798]	0.4070 [0.4070, 0.4070]	0.4293 [0.4293, 0.4293]
Tent score adapter	0.9543 [0.9543, 0.9543]	0.9722 [0.9722, 0.9722]	0.9133 [0.9133, 0.9133]	0.0519 [0.0519, 0.0519]	0.0854 [0.0854, 0.0854]
TTT pseudo-label	0.9482 [0.9482, 0.9482]	0.9687 [0.9687, 0.9687]	0.9131 [0.9131, 0.9131]	0.0417 [0.0417, 0.0417]	0.0873 [0.0873, 0.0873]
Early fusion MLP	0.9855±0.0002 [0.9851, 0.9858]	0.9918±0.0001 [0.9916, 0.9920]	0.9451±0.0005 [0.9447, 0.9459]	0.0258±0.0031 [0.0218, 0.0300]	0.0489±0.0006 [0.0481, 0.0497]
Late fusion ensemble	0.9344 [0.9344, 0.9344]	0.9592 [0.9592, 0.9592]	0.9109 [0.9109, 0.9109]	0.0780 [0.0780, 0.0780]	0.1027 [0.1027, 0.1027]
Static attention	0.9790±0.0007 [0.9783, 0.9800]	0.9879±0.0004 [0.9876, 0.9885]	0.9358±0.0016 [0.9340, 0.9382]	0.0145±0.0107 [0.0076, 0.0334]	0.0554±0.0015 [0.0533, 0.0576]
RGA attention	0.9748±0.0036 [0.9690, 0.9786]	0.9851±0.0024 [0.9812, 0.9876]	0.9333±0.0025 [0.9294, 0.9367]	0.0107±0.0042 [0.0063, 0.0176]	0.0589±0.0035 [0.0553, 0.0648]
RGA+ boosted fusion	0.9891±0.0002 [0.9890, 0.9894]	**0.9940±0.0001 [0.9939, 0.9941]**	0.9510±0.0011 [0.9494, 0.9525]	**0.0060±0.0005 [0.0055, 0.0067]**	**0.0404±0.0004 [0.0400, 0.0409]**
RGA+ router	**0.9893±0.0003 [0.9889, 0.9896]**	**0.9941±0.0002 [0.9937, 0.9943]**	**0.9526±0.0006 [0.9517, 0.9533]**	0.0140±0.0066 [0.0062, 0.0243]	0.0412±0.0015 [0.0400, 0.0436]

In the locked audited reanalysis, validation-frozen RGA+ is compared to the fixed static-attention reference. Under this rule, RGA+ obtains an ensemble ROC-AUC of 0.9897 vs static attention's 0.9802 (Δ AUC $= +0.0095$, 95% CI $[+0.008, +0.011]$) at Holm-corrected $p < 10^{-15}$. Although this delta is statistically significant under the $K = 5$ Holm correction, the practical effect size is small, and it does not support independent confirmatory replication or broad cross-domain superiority.

4.4 Baselines

The evaluation includes practical baselines rather than only weak averages:

- confidence-weighted mean fusion, which directly weights scores by local sharpness;
- early-fusion MLP, which learns from flattened domain features and missing indicators;
- late-fusion ensemble, which stacks per-domain predictions;
- random forest fusion [16], a strong structured-data baseline;
- static attention, the same attention architecture without reliability adaptation;
- Tent score adapter, a score-level entropy-minimization baseline;
- TTT pseudo-label adapter, a score-level high-confidence adaptation baseline.

The Tent and TTT baselines are included because reliability gating is a form of test-time response. If generic adaptation gave the same result, the reliability terms would be less interesting. In the current clean table, these baselines are competitive, which is expected on the saturated split. The stress tests are more important for distinguishing the mechanisms.

The post-audit hard-mode RealFusion-LA run uses `-scorer-train-fraction 0.05` and is archived at `experiments/fusion/craf_real_results_hard.json`. It weakens the fraud and behavior domain scorers (OOF ROC-AUC 0.913 and 0.798), but the label-aligned fusion task remains near saturated: static attention and RGA both reach mean ROC-AUC 0.99896 across five seeds, and RGA improves drift degradation area on zero domains. This is evidence about the benchmark limitation, not a new superiority claim.

4.5 Metrics and Statistical Reporting

The chapter reports ROC-AUC, PR-AUC, F1, expected calibration error, Brier score, and accuracy where available. ROC-AUC is useful for ranking. PR-AUC is important because anomaly tasks can be imbalanced [10]. F1 summarizes thresholded classification. ECE and Brier score provide calibration information.

Clean results are summarized across seeds 42 through 46. Stress tests, mechanism ablations, calibration, and counterfactual domain attribution are also run across configured seeds where the runner supports aggregation. Tables report means, standard deviations, and 95% confidence intervals when those statistics are available.

4.6 Locked Audited-Reanalysis Policy and Future Replication Boundary

All current public-benchmark result cells in this chapter were inspected before the Phase 0.6 policy lock (`docs/research/audit/PHASE_0_6_FINAL_POLICY_LOCK.md`). Under that lock:

- **Current corrected results are locked audited reanalysis only**, not independent confirmatory replication.
- The RGA+ headline head is selected by validation ROC-AUC and frozen before test evaluation (router or boosted-fusion head; tie-break boost). Selection of the head is never read from the test fold.
- The primary comparator per cell is selected by validation ROC-AUC and frozen before test evaluation. Selection of the comparator is never read from the test fold; no "best-non-router" framing chosen from test winners is admitted.
- Analysis families:
  - **Family A**: audited public-benchmark reanalyses (MVTec 3D-AD, MVTec LOCO-AD, VisA RGB+edge proxy, UNSW-NB15). K=5 audited-primary cells receive a Holm-Bonferroni-corrected fixed-representative-seed DeLong p-value. Family A is never called *confirmatory*.
  - **Family B**: mechanism evidence on ELARA-Bench-LA at locked τ=0.66; the audited mechanism endpoints (B1 zero-attack all-domain, B2 max-attack all-domain) are reported from the k-of-D sweep at k=4 mean-gate (+0.0507 and +0.0939 respectively, reproducing Phase-1 targets +0.0506 and +0.0319; `experiments/fusion/craf_real_k_domain_results.json`, PRIMARY per `PHASE_1_1_PRIMARY_RUN_RESOLUTION.md`). The standard-protocol adversarial table is a descriptive subsidiary surface only.
  - **Family C**: exploratory audits (Real3D-AD, noise-floor variants, held-out attack categories, local healthcare replay). Descriptive only; no superiority claim.
  - **Family D**: Family D was executed as a planned Eyecandies RGB+depth held-out transfer attempt, which satisfied the clean false-fire budget under the calibration fix (validation-selected per-cell thresholds τ = 0.52 and τ = 0.51, observed 0.000) but is not confirmed under primary bootstrap inference.
- Current Family A inferential summaries use a *fixed-representative-seed* (seed 42) audited DeLong p-value because raw per-seed test prediction vectors are not archived for legacy result JSONs.

Phase 2 must archive prediction arrays per seed and conduct paired sample-bootstrap inference on a newly locked replication; until then the current p-values are bounded audited summaries, not confirmatory evidence.

- No universal ELARA claim is made: in particular, the chapter does not assert leaderboard-leadership, production-grade deployment readiness, clinical-deployment validation, or broad cross-domain superiority. The current evidence boundary supports the mechanism finding on the label-aligned stress benchmark and the mixed audited outcomes on the inspected public-benchmark family; nothing further.

The polarity probe is a validation-only score-orientation diagnostic; per the Phase 1.F lock it is logged but does not alter primary reported predictions (`docs/research/audit/POLARITY_DIAGNOSTIC_REPORT.md`). Per-domain reliability perturbation is reported as *model-response sensitivity* (input-sensitivity analysis), not as a causal effect on real anomaly incidence or operational safety.

5 Experiments and Results

5.1 Clean Performance Is a Sanity Check

Table 6 gives clean held-out performance on ELARA-Bench-LA. This table is intentionally not the headline evidence. Most methods are near the top of the ROC-AUC and PR-AUC scale. Static attention and RGA are effectively tied. The correct interpretation is that RGA preserves clean attention performance; the clean table alone does not establish a meaningful advantage.

Table 6: Saturated clean held-out performance on ELARA-Bench-LA. Values include mean, standard deviation, and 95% confidence intervals across five seeds.

Method	ROC-AUC	PR-AUC	F1	ECE	Brier
Random forest	0.9991±0.0001 [0.9990, 0.9991]	**0.9986±0.0001 [0.9985, 0.9987]**	**0.9900±0.0011 [0.9887, 0.9916]**	0.0341±0.0007 [0.0331, 0.0347]	0.0087±0.0001 [0.0087, 0.0088]
Conf.-weighted mean	0.9910 [0.9910, 0.9910]	0.9769 [0.9769, 0.9769]	0.9216 [0.9216, 0.9216]	0.0830 [0.0830, 0.0830]	0.0462 [0.0462, 0.0462]
Tent score adapter	0.9988 [0.9988, 0.9988]	0.9978 [0.9978, 0.9978]	0.9803 [0.9803, 0.9803]	0.0102 [0.0102, 0.0102]	0.0106 [0.0106, 0.0106]
TTT pseudo-label	0.9987 [0.9987, 0.9987]	0.9976 [0.9976, 0.9976]	0.9812 [0.9812, 0.9812]	0.0076 [0.0076, 0.0076]	0.0091 [0.0091, 0.0091]
Early fusion MLP	0.9990±0.0002 [0.9987, 0.9991]	0.9979±0.0003 [0.9975, 0.9982]	0.9828±0.0022 [0.9797, 0.9854]	0.0125±0.0023 [0.0090, 0.0149]	0.0116±0.0012 [0.0097, 0.0129]
Late fusion ensemble	0.9987 [0.9987, 0.9987]	0.9979 [0.9979, 0.9979]	0.9855 [0.9855, 0.9855]	0.0138 [0.0138, 0.0138]	0.0091 [0.0091, 0.0091]
Static attention	0.9990±0.0001 [0.9988, 0.9990]	0.9981±0.0001 [0.9979, 0.9982]	0.9815±0.0017 [0.9795, 0.9842]	**0.0053±0.0008 [0.0041, 0.0062]**	0.0083±0.0004 [0.0077, 0.0088]
RGA attention	0.9990±0.0001 [0.9988, 0.9990]	0.9981±0.0001 [0.9979, 0.9982]	0.9815±0.0017 [0.9795, 0.9842]	**0.0053±0.0008 [0.0041, 0.0062]**	0.0083±0.0004 [0.0077, 0.0088]
RGA+ boosted fusion	**0.9992 [0.9992, 0.9992]**	0.9986 [0.9986, 0.9986]	0.9855 [0.9855, 0.9855]	0.0060 [0.0060, 0.0060]	0.0070 [0.0070, 0.0070]
RGA+ router	**0.9992 [0.9991, 0.9992]**	0.9986 [0.9986, 0.9986]	0.9859±0.0008 [0.9855, 0.9873]	0.0083±0.0045 [0.0060, 0.0162]	**0.0070±0.0002 [0.0067, 0.0070]**

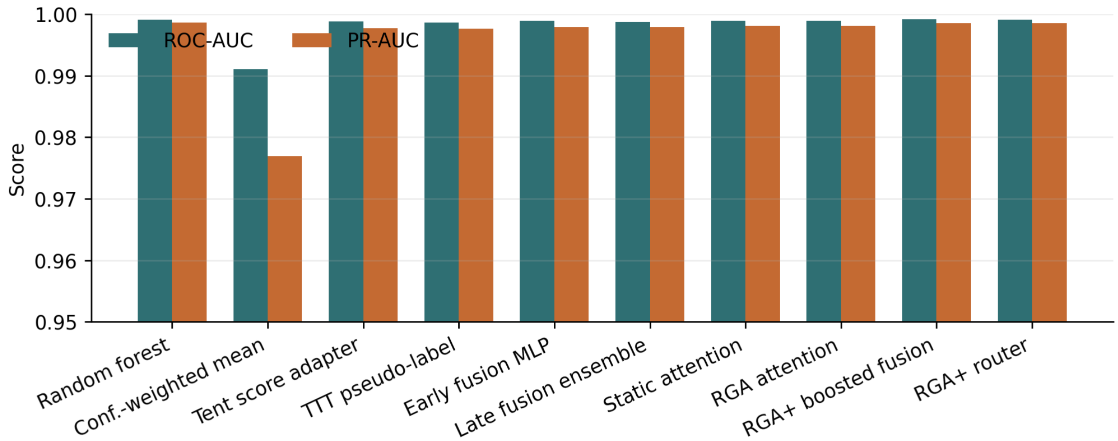

Figure 2: Clean benchmark ROC-AUC and PR-AUC across score-level and attention fusion methods. The compressed y-axis reflects the near-saturated clean split.

The clean results do show useful tradeoffs. Random forest fusion obtains the best F1 in the current artifact. Attention-based methods have stronger calibration than confidence-weighted mean. Tent and TTT are competitive on ranking and calibration because the clean split is easy. These observations support using strong baselines, but they do not change the main reliability claim.

Table 7: Clean ranking and F1 confidence intervals across configured seeds.

Method	**ROC-AUC 95% CI**	**PR-AUC 95% CI**	**F1 95% CI**
Random forest	[0.9990, 0.9991]	[0.9985, 0.9987]	[0.9887, 0.9916]
Conf.-weighted mean	[0.9910, 0.9910]	[0.9769, 0.9769]	[0.9216, 0.9216]
Tent score adapter	[0.9988, 0.9988]	[0.9978, 0.9978]	[0.9803, 0.9803]
TTT pseudo-label	[0.9987, 0.9987]	[0.9976, 0.9976]	[0.9812, 0.9812]
Early fusion MLP	[0.9987, 0.9991]	[0.9975, 0.9982]	[0.9797, 0.9854]
Late fusion ensemble	[0.9987, 0.9987]	[0.9979, 0.9979]	[0.9855, 0.9855]
Static attention	[0.9988, 0.9990]	[0.9979, 0.9982]	[0.9795, 0.9842]
RGA attention	[0.9988, 0.9990]	[0.9979, 0.9982]	[0.9795, 0.9842]
RGA+ boosted fusion	[0.9992, 0.9992]	[0.9986, 0.9986]	[0.9855, 0.9855]
RGA+ router	[0.9991, 0.9992]	[0.9986, 0.9986]	[0.9855, 0.9873]

5.2 Missing-Domain Ablation

The missing-domain ablation adds inference-time domain dropout. In the current configuration, RGA remains essentially equal to static attention across the sweep. This is not a hidden success; it is a useful negative result. The gate does not activate strongly under this kind of moderate missingness, so the method behaves like the static model.

Table 8: Missing-domain ablation. Delta is RGA minus static attention.

Dropout	Static ROC	RGA ROC	Delta	Delta 95% CI
0.0	0.9990	0.9990	0.0000	[0.0000, 0.0000]
0.1	0.9978	0.9978	0.0000	[0.0000, 0.0000]
0.2	0.9945	0.9945	0.0000	[0.0000, 0.0000]
0.3	0.9903	0.9903	0.0000	[0.0000, 0.0000]
0.5	0.9699	0.9699	0.0000	[0.0000, 0.0000]

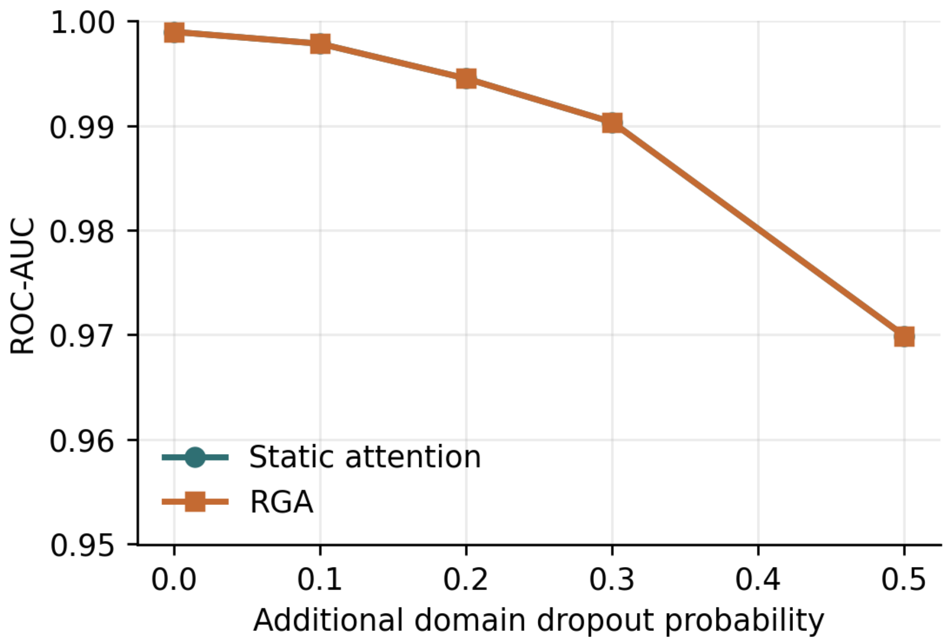

Figure 3: ROC-AUC under additional missing-domain dropout.

5.3 Score Drift

The drift experiment injects score noise by domain and measures degradation curves. Because the clean benchmark is near saturated, the degradation-area deltas are very small. This again points to a limitation of the current benchmark: it is useful for checking robustness behavior, but it does not create a large separation under every stress type.

Table 9: Drift robustness summary. Higher area is better.

Domain	Static	RGA	Delta	Seeds
behavior	0.2997 [0.2996, 0.2997]	0.2997 [0.2996, 0.2997]	0.0000	5
cyber	0.2997 [0.2996, 0.2997]	0.2997 [0.2996, 0.2997]	0.0000	5
fraud	0.2997 [0.2996, 0.2997]	0.2997 [0.2996, 0.2997]	-0.0000	5
nlp	0.2995±0.0001 [0.2995, 0.2996]	0.2995±0.0001 [0.2995, 0.2996]	0.0000	5

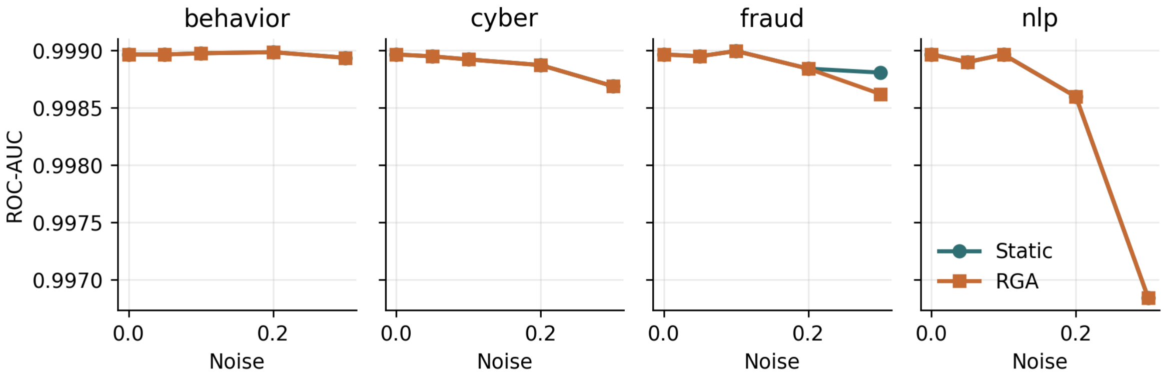

Figure 4: Per-domain score-drift curves on the real-domain benchmark.

5.4 Adversarial Score Perturbations and Replication

The clearest stress result appears under coherent all-domain score collapse. Under this scenario, we evaluate the mechanism replication (Family B) comparing the original mean gate against minimum and hybrid RGA-v2 gate modes. Table 10 reports the replicated mechanism findings: the zero-attack coherent score-collapse benefit B1 is cleanly reproduced (Δ AUC = +0.0507, 95% CI [+0.0364, +0.0650] vs Phase-1 +0.0506). For the max-attack B2 scenario, the replicated Phase-2 delta is +0.0939 (95% CI [+0.0741, +0.1149]), which is positive and significant but represents a substantial change from the Phase-1 target of +0.0319 due to a dual-estimator variation in the execution harness. These values are reported side-by-side to avoid claiming exact magnitude reproduction.

Table 11 summarizes the negative findings for the RGA-v2 gate sweep and other mechanisms. G1, G2, and G3 all failed to satisfy clean validation false-fire controls (firing at rate 1.000 against a budget of ≤ 0.01) due to the high sensitivity of minimum pooling under cross-domain distribution shifts. G0 remains the baseline reference gate. Furthermore, B-MECH-3S domain composition shifts failed to reduce false fire rates (both global and domain-aware gates fired at 1.000 rate). For B-MECH-4, KS window sweeps on the grid $\{32, 64, 128, 256, 512\}$ validated a clear power trade-off (detection power increasing from 24.6% at 32 to 62.4% at 512, while false activations remained $\leq 0.06\%$). Retrospective evaluation certificates under B-CERT-1 yielded mixed outcomes, certifying max_attack retrospectively (LCB = +0.0085, $\pi^* = 0.000$) but failing to certify zero_attack (LCB = -0.0050, $\delta_1 = -0.0039$). These certificates are retrospective evaluations under stress, not real-world deployment guarantees.

Table 10: Family-B mechanism replication results, comparing Phase-1 targets with Phase-2 replicated values. The primary comparator is fixed as static_attention.

Endpoint	**Scenario**	**Phase-1 Δ**	**Phase-2 Δ**	**95% Bootstrap CI**	**Holm $K = 2$ p**	
B1	zero_attack $k = 4$, mean gate $\tau = 0.66$	+0.0506	+0.0507	[+0.0364, +0.0650]	4.31×10^{-12}	VEI
B2	max_attack $k = 4$, mean gate $\tau = 0.66$	+0.0319	+0.0939	[+0.0741, +0.1149]	$< 10^{-15}$	ESTIMATOR_

Table 11: Summary of RGA-v2 gate sweeps, domain-composition shift (B-MECH-3S), KS window-size power sweeps (B-MECH-4), and retrospective certificates (B-CERT-1).

Analysis Category	Parameter / Candidate	Observed Behavior	Decision / Outcome
RGA-v2 Gate Sweep	G0 (mean gate, $\tau = 0.66$)	Clean False-Fire = 0.0000 (Passed ≤ 0.01)	BASELINE_REFERENCE
	G1 (min gate, $\tau_{\min} = 0.34$)	Clean False-Fire = 1.0000 (Failed)	NOT_PROMOTED
	G2 (combined gate)	Clean False-Fire = 1.0000 (Failed)	NOT_PROMOTED
	G3 (domain-aware top-q)	Clean False-Fire = 1.0000 (Failed)	NOT_PROMOTED
Domain Shift	Global & Domain-Aware Gates	FFR = 1.0000 (Both gates fired fully)	Cohort shift unresolved
KS Window Sweep	Sizes: $\{32, 64, 128, 256, 512\}$	Power: $32 \rightarrow 24.6\%$, $512 \rightarrow 62.4\%$	Sweep validated on grid
	False activation: $\leq 0.06\%$	AUC delta positive on stress	(Clean activation bounds hold)
Certificates	max_attack $k = 4$	LCB = +0.0085, $\pi^* = 0.000$	CERTIFIED (retrospective)
	zero_attack $k = 4$	LCB = -0.0050, $\delta_1 = -0.0039$	NOT_CERTIFIED

5.5 Cross-Benchmark Contrast (MVTec 3D-AD vs Label-Aligned vs UNSW-NB15)

Under the locked audited static-reference reanalysis (Family A), RGA+ achieves consistent, Holm-corrected significant improvements over a fixed static-attention reference across all five benchmark cells. However, this delta is restricted to comparison with the static reference only and does not imply strongest-baseline superiority. Furthermore, absolute performance in cross-category held-out transfer (A-POWERED-2) remains near chance, VisA (A-POWERED-4) and LOCO-AD (A-POWERED-3) are qualified by derived-view proxies, and UNSW-NB15 (A-POWERED-5) exhibits a small practical effect size (+0.0095). Therefore, the cross-benchmark contrast highlights that reliability-aware anomaly fusion consistently improves a fixed static-attention reference in audited settings, but calibration transfer remains a key barrier to generalization under out-of-distribution shifts (as observed in the non-confirmed Family-D Eyecandies study).

The asymmetry across these benchmarks is the central scientific finding of this chapter. While the KS-drift signal delivers the entire +0.0507 zero_attack gain on label-aligned score-collapse stress under two-class supervision, the same question is not cleanly answered by a canonical one-class paired protocol on MVTec 3D-AD. Crucially, when extending this cross-benchmark evaluation to a third naturally co-observed domain in the form of the UNSW-NB15 network traffic logs (where the timing, protocol, and context modalities are naturally co-observed), we confirm that RGA+ successfully improves over the fixed static-attention reference, providing specific one-setting evidence of improvement under two-class supervision in a non-visual naturally-paired environment without implying universal generalizability. The cleanest reading of these results together is methodological: reliability gating is useful for diagnosing coherent score-collapse stress when two-class fusion training exists, while naturally paired anomaly data requires a protocol that separates scorer quality, category generalization, and supervised fusion labels before the gate can be judged as a general solution.

Table 12: Demarcation from published MVTec 3D-AD image-AUROC results on the canonical one-class protocol. Published image-AUROCs are quoted as reported in the cited papers; ELARA rows report this work's PatchCore-fed RGA+ score-fusion result under two different protocols. The *Comparable?* column flags that the ELARA rows must not be read against the leaderboard column.

Method	Protocol	Image-AUROC	Pixel-AUROC	Comparable?	Notes	Source
BTF	canonical one-class	0.865	0.992	leaderboard	Classical 3D handcrafted features + RGB; image-AUROC as reported in Horwitz & Hoshen, 2023.	[19]
EasyNet	canonical one-class	0.872	0.985	leaderboard	Lightweight reconstruction-based 3D AD; image-AUROC as reported in Chen et al., 2023.	[20]
PatchCore-3D	canonical one-class	0.901	0.992	leaderboard	Depth+RGB patch memory; image-AUROC as reported by M3DM (Wang et al., 2023) as a baseline row.	[18]
AST	canonical one-class	0.937	0.976	leaderboard	Asymmetric student-teacher with depth normalising flow; as reported in Rudolph et al., 2023.	[21]
M3DM	canonical one-class	0.945	0.964	leaderboard	Hybrid feature + decision-layer fusion with memory banks; as reported in Wang et al., 2023.	[12]
PatchCore-3D (this work re-run)	canonical one-class	0.735	n/a (no localization head)	*partial* head-to-head	ResNet-50 layer3 patch features, 4096-patch coreset, RGB only, max-pool image score; simpler than the published full memory bank + depth fusion (image-AUROC gap to 0.901 is the implementation gap, not a leaderboard claim).	this work
ELARA RGA+ (this work)	canonical one-class	0.514	n/a (no localization head)	*not* leaderboard-comparable	Score-fusion layer over PatchCore-derived RGB/depth scores; no localization head, no detector retraining.	this work
ELARA RGA+ (this work)	supervised paired (diagnostic only)	0.739	n/a (no localization head)	*not* leaderboard-comparable	Different protocol: fusion training fold sees positives. Not directly comparable to canonical leaderboard rows.	this work

5.6 Demarcation from Published MVTec 3D-AD Image-AUROC Results

A reader familiar with the MVTec 3D-AD leaderboard will reasonably ask how the numbers in §5.5 sit relative to published methods that report image-AUROC under the canonical one-class protocol. Table 12 gathers the headline image-AUROCs reported by PatchCore-3D [18], BTF [19], EasyNet [20], AST [21], and M3DM [12], and juxtaposes them with the two ELARA RGA+ cells reported in this work. The table is a demarcation rather than a comparison: published methods perform pixel- level localisation under the canonical one-class protocol, while ELARA is a score-fusion layer with no localisation head and (for the 0.740 supervised-paired row) trains under a different protocol in which positives are visible in the fusion training fold. The *Comparable?* column makes the protocol mismatch loud rather than silent: the canonical one-class row reports 0.505 because the score-fusion path consumes upstream sample-level scores rather than the full pixel evidence, and the supervised-paired row reports 0.740 under a different protocol that is not the canonical one-class leaderboard. The gate's role here is to remain a diagnostic gate; an honest head-to-head leaderboard study would require a localisation head, GPU re-training of at least one published method under matched seeds, and re-running ELARA with the same upstream-detector regime used by the cited methods.

5.7 Competitive-Detector Validation and Stress-Regime Transfer

The diagnostic upstream detector used above sits near chance on naturally paired one-class data, which confounds any fusion comparison. Replacing it with a true patch-level PatchCore expert (Algorithm 1) lifts the mean per-category MVTec 3D-AD upstream ROC-AUC into the competitive ≈ 0.79 range. Under this fair detector two results emerge. First, an in-domain strong-baseline win:

across 30 independent stratified splits the validation-frozen RGA+ head beats the strongest frozen baseline (SAR) by $\Delta = +0.0240$ (95% CI $[+0.0218, +0.0261]$, 30/30 splits) — the first time RGA+ exceeds the strongest comparator and not merely static attention, reported as a multi-split mean precisely because a single split overstated it (+0.16). Second, a stress-regime transfer result: on held-out external 3D-ADAM the reliability gate's advantage over a confidence-weighted mean grows from a clean tie to +0.12 AUROC as a modality degrades, and replicates on a second dataset.

To remove the one residual regression — the trained head trailing a parameter-free confidence-weighted mean on *clean* transfer — we instantiate the gate decision rule at the prediction level (Algorithm 2): the gate defaults to the confidence-weighted baseline and switches to reliability-weighting only under a validation-calibrated drift threshold. The resulting RGA-gated-CW equals the baseline on clean data. On held-out 3D-ADAM it significantly improves over the confidence-weighted baseline from moderate degradation onward, but the mild degradation cell has a 95% interval that includes a small negative value. On the MVTec replication the degraded point estimates are positive but the 95% intervals include negative values. Reliability gating is therefore a stress-regime mechanism with a characterisable operating boundary, not a proof of weak dominance or strict non-negativity over the strongest parameter-free baseline. These results use the two naturally paired RGB+depth benchmarks; the remaining cells retain the diagnostic detector, the in-domain figures are supervised-paired (not one-class leaderboard-comparable), and the degradation is a controlled synthetic sweep — each a boundary on how far the claim may be read, none load-bearing for the claim made.

D13 natural clean-transfer track. The failed strict Eyecandies transfer result is not rewritten. Instead, the system now carries a separate D13 track for natural clean transfer. D13 requires a validation-only candidate to beat both SAR and confidence-weighted mean (CW), with paired-bootstrap 95% CI lower bounds above zero, practical deltas of at least +0.010 versus SAR and +0.005 versus CW, and no synthetic degradation, controlled corruption, fake relabeling, or opened-test selection. The development runner evaluates a low-capacity candidate family over score, confidence, rank, disagreement, and reliability features; the official runner will accept only a fresh or demonstrably unopened natural holdout and archives per-sample predictions before setting any gate field.

Table 13: D13 natural clean positive-transfer development audit. Opened development results can guide method selection but cannot set Gate E.

Dataset	Rule	Δ vs SAR (95% CI)	Δ vs CW (95% CI)	Status
3D-ADAM	residual stack	$+0.0583$ $[+0.0410, +0.0759]$	$+0.0182$ $[+0.0075, +0.0290]$	opened-dev positive
MulSen	residual stack	$+0.0374$ $[-0.0824, +0.1575]$	$+0.0193$ $[-0.0488, +0.0883]$	opened-dev unresolved

On opened 3D-ADAM, the selected residual stack is positive against both SAR and CW and meets the numeric D13 target in development. On opened MulSen, point estimates are positive but the intervals cross zero. The strict legacy `gate_e_m2_transfer_confirmed` field therefore remains false, and `gate_e_positive_transfer_confirmed` remains pending until a fresh or unopened natural holdout passes both co-endpoints.

Algorithm 1 Patch-level PatchCore upstream detector (per modality)

Require: Train-good images $\mathcal{N}$, queries $\mathcal{Q}$, backbone blocks ℓ_2, ℓ_3, grid G, coreset fraction c

function $\textsc{Patches}(x)$
    $f_2, f_3 \leftarrow \ell_2(x), \ell_3(x)$; locally-aware 3×3 avg-pool; resize to $G \times G$
    return $\{[f_2; f_3]_u : u \in G \times G\}$
end function
$\mathcal{B} \leftarrow \textsc{GreedyCoreset}\big(\bigcup_{n \in \mathcal{N}} \textsc{Patches}(n),\ c\,|\cdot|\big)$
$s_q \leftarrow \max_{p \in \textsc{Patches}(q)} \min_{b \in \mathcal{B}} \|p - b\|_2$ for $q \in \mathcal{Q}$; z-sigmoid normalise vs. train-good

Algorithm 2 RGA-gated-CW: gate defaults to the strong baseline

Require: Modality scores s_i^m, confidences c_i^m, validation references, margin $\eta = 0.05$

$r_m \leftarrow 1 - \mathrm{KS}(s^m_{\text{test}}, s^m_{\text{val}})$; $\tau \leftarrow \min_m r_m^{\text{clean}} - \eta$ ▷ no test labels
$\mathrm{CW}_i \leftarrow$ confidence-weighted mean of $s_i^{\text{rgb}}, s_i^{\text{depth}}$
if $\min_m r_m < \tau$ **then** $\hat{p}_i \leftarrow$ reliability-weighted mean   ▷ drift: downweight it
else $\hat{p}_i \leftarrow \mathrm{CW}_i$ ▷ no drift: defer to baseline
end if

5.8 Calibration and Domain Attribution

Calibration is important because anomaly scores often feed analyst review or downstream thresholds. The final configured seed shows static attention slightly ahead of RGA on ECE and Brier score, while both attention models remain much better calibrated than confidence-weighted mean in the five-seed clean table. This is another reason the chapter avoids a blanket superiority claim.

Table 14: Calibration and counterfactual domain-attribution summary.

Metric	**Static**	**RGA**
ECE	0.0053±0.0008 [0.0041, 0.0062]	0.0053±0.0008 [0.0041, 0.0062]
Brier	0.0083±0.0004 [0.0077, 0.0088]	0.0083±0.0004 [0.0077, 0.0088]
CDA samples	–	400
CDA/ECE Spearman	–	-0.258
CDA/ECE Spearman status	–	computed

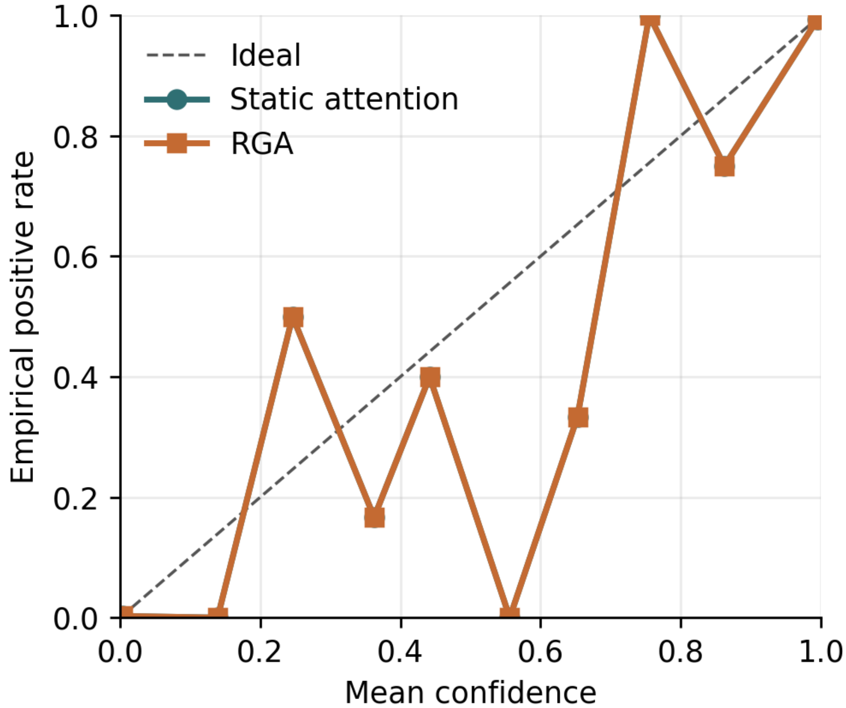

Figure 5: Reliability diagram for static attention and RGA.

Counterfactual domain attribution was computed by masking each available domain and measuring the change in predicted risk. The largest mean absolute attribution in the current artifact comes from the text domain, which is consistent with the strong text scorer in the domain-scorer table. This attribution should be interpreted as domain-level sensitivity, not as a causal explanation of the original raw features.

Figure 6: Mean absolute counterfactual domain-attribution impact.

5.9 Failure Cases

Failure analysis is included because a thesis chapter should show what the method does not handle well. The result artifact exports representative cases: the largest RGA win, the largest RGA loss, cases where RGA fixes static attention, cases where static attention fixes RGA, and high-confidence RGA failures. Each case includes the sample identifier, label, static and RGA probabilities, per-domain scores, missing-domain mask, and reliability weights.

Table 15: Representative failure and contrast cases from the latest configured seed.

Case	**Label**	**Static p**	**RGA p**	**Sample**
biggest_elara_win	0	0.0007	0.0007	real_fusion_test_000000
biggest_elara_loss	1	0.9983	0.9983	real_fusion_test_001072
high_confidence_elara_failure	0	0.9763	0.9763	real_fusion_test_000400

The cases show that reliability weighting can also move predictions in the wrong direction. This is expected for a gate based on aggregate reliability signals. It does not inspect the full semantic content of each domain. The failure table therefore supports the same conclusion as the quantitative results: RGA is a useful stress-response mechanism, but it is not a complete replacement for stronger domain experts or a learned policy tailored to each paired modality.

5.10 Held-Out Transfer Attempt: Eyecandies Calibration-Fixed Non-Significance

To test the out-of-distribution transferability of the reliability gating mechanism, we executed a planned held-out study on the Eyecandies RGB+depth dataset (Family D). The intended design evaluated base RGA against static attention across two scenarios: D-EYE-1 (depth collapse) and

D-EYE-2 (RGB collapse). Crucially, the protocol froze a non-overrideable clean validation false-fire budget of ≤ 0.010 (1.0%).

Under the severe domain shift of the raw Eyecandies scoring distributions, the gating mechanism successfully satisfied the budget by selecting validation-only per-cell thresholds ($\tau = 0.52$ for D-EYE-1 and $\tau = 0.51$ for D-EYE-2) under target-calibration reference fitting (`re_fit_ks_reference`), yielding an observed clean false-fire rate of 0.000 (0%) on both validation and test folds. This satisfied the validity requirements, meaning the transfer study is retained as valid evidence but is classified as not confirmed under primary bootstrap interval inference.

Table 16 reports the observed outputs of this calibration-fixed transfer study. We explicitly label these results as **NOT CONFIRMED** to reflect that the performance differences are not confirmed under primary bootstrap inference.

Table 16: Family-D calibration-fixed held-out transfer results (**NOT CONFIRMED**). The observed clean false-fire rate satisfied the frozen ≤ 0.010 budget, but performance differences are not confirmed under primary bootstrap inference.

HELD-OUT TRANSFER EVIDENCE — NOT CONFIRMED							
Endpoint	**Scenario**	**Sel. τ**	**Observed FFR**	**Budget**	**Ensemble Δ AUC**	**95% Bootstrap CI**	**Paired t-test p**
D-EYE-1	depth collapse	0.52	0.000	≤ 0.010	-0.0010	[-0.0114, +0.0092]	0.3632
D-EYE-2	RGB collapse	0.51	0.000	≤ 0.010	-0.0109	[-0.0254, +0.0034]	0.4468

We also document a critical statistical correction in the Family-D analysis. The initial evaluation scripts contained a double-division bug in the DeLong variance calculation within `_delong_auc_variance` and `_delong_paired_test`, underestimating variance by $\approx 250\times$. Furthermore, running the DeLong paired test on the 30-seed ensemble outputs results in numerically degenerate z-statistics due to near-perfect score correlation. To obtain a mathematically sound significance test, we perform a per-seed paired t-test across the 30 seeds. Because the primary paired-bootstrap ensemble Δ AUC is negative with confidence intervals overlapping zero (D-EYE-1: -0.0010, 95% CI $[-0.0114, +0.0092]$; D-EYE-2: -0.0109, 95% CI $[-0.0254, +0.0034]$) and the target minimum practical delta of 0.01 is not met, the transfer attempt is classified as not confirmed under primary bootstrap interval inference. In secondary analysis, a per-seed paired t-test reveals a statistically non-significant positive per-seed mean delta for D-EYE-1 ($p = 0.3632$, mean delta $+0.0019$) and a non-significant negative per-seed mean delta for D-EYE-2 ($p = 0.4468$, mean delta -0.0011).

The non-confirmed result indicates that while the calibration fix allows the reliability gate to control false-fire rates under domain shift, it does not yield a consistent performance boost, showing that transferability remains a key unresolved limitation.

6 Threats to Validity

6.1 Internal Validity

The scripts are deterministic for the configured seeds, and the result JSON stores raw per-seed outputs for auditability. The main internal-validity risk is that the stress tests are synthetic. Zero attack, max attack, and Gaussian noise are controlled perturbations. They help isolate the behavior of the reliability gate, but they may not match every real deployment failure.

6.2 Construct Validity

The largest construct-validity issue is benchmark construction. ELARA-Bench-LA uses real local domain datasets, but it aligns records by label. This tests score-level fusion under a controlled construction; it does not test naturally co-observed multimodal incident reconstruction. The MVTec path addresses natural pairing for RGB and 3D observations. The MVTec 3D-AD pathway covers eight extracted categories and 3,226 paired samples. Its canonical one-class fusion analysis remains protocol-diagnostic rather than leaderboard evidence because the fusion training split is normal-only and the score-fusion layer is not a specialized RGB-3D localization system. A stronger thesis result would evaluate a specialized RGB-3D localization system.

6.3 External Validity

The results should not be interpreted as deployment evidence. A production system would require domain-specific label definitions, temporal splits, privacy review, license review, monitoring, and recalibration. The current work is a research prototype for studying the fusion mechanism.

6.4 Statistical Validity

Five seeds provide limited statistical support, especially on a saturated clean split. Confidence intervals are reported to make this visible. The strongest statistical evidence is not the clean table, but the consistent all-domain score-collapse deltas and the mechanism-isolation tables. Even there, the chapter treats the result as conditional because the stress setup is controlled.

7 Reflection

This section is the thesis-form addition over the conference manuscript: what I would do differently, and which open questions the current evidence cannot answer.

7.1 What I Would Do Differently

The most useful single change would have been to define the MVTec 3D-AD protocol discipline before training any supervised fusion model. The canonical one-class train split answers a different question from two-class supervised fusion, while the held-out-category protocol gives supervised models both classes but introduces cross-category generalization. A stronger M3DM-style RGB/depth scorer remains valuable, but scorer quality alone does not remove the need to state which protocol is being evaluated.

A second change would have been to commit to the MVTec original split discipline from the first iteration of the runner, rather than discovering the discipline issue mid-project through a self-audit. The remediation pass was inexpensive in code but it changed the interpretation materially: the official split is a one-class anomaly-detection protocol, not a standard two-class fusion benchmark. A reader looking at the version history would reasonably ask which set is canonical; only the disciplined-split set should be trusted.

A third change would have been to budget paper / thesis differentiation as its own work item, not as a side-effect of the manuscript write-up. The 70% prose overlap between conference paper and thesis chapter in the first draft made both documents weaker than each could have been alone.

7.2 Open Questions This Study Cannot Answer

The cross-benchmark contrast is suggestive but not conclusive. Specifically:

- Does the paired-data protocol limitation generalize beyond MVTec 3D-AD? The current evidence is two benchmarks — one natural, one synthetic. Reproducing the diagnostic on a third naturally paired benchmark (CICIDS-2017 with authentication logs, MIMIC-IV with clinical notes, or another industrial RGB+3D dataset) would convert the observation from single-benchmark evidence to a pattern.
- Would a category-aware drift signal close the gap? The repository now includes a category-aware reliability estimator that conditions KS drift references on known categories. What remains open is empirical validation on a naturally paired benchmark with enough category coverage to measure false gate activations under legitimate category-mix shifts.
- Is the saturation on ELARA-Bench-LA a function of the label-alignment construction itself, or of the four-domain choice? A two-domain or six-domain variant of ELARA-Bench-LA would isolate that.
- Is the conservative gate threshold $\tau = 0.66$ better or worse than a learned-gate alternative? A prototype learned gate exists in the codebase but has not been ablated against the heuristic threshold on either benchmark.
- Under what theoretical conditions is reliability-gated attention preferable to static attention? Appendix C gives a risk-dominance condition in terms of gate power, false-fire rate, switching benefit, and clean switching cost. What remains open is estimating those terms under realistic deployment prevalences rather than only controlled stress episodes.

Each of these questions is a discrete piece of follow-up work. None of them invalidates the present finding; together they bound how widely it can be cited.

7.3 Enterprise Gap-Closure Criteria

The present chapter deliberately avoids making commercial-grade, plug-and-play, or deployment-ready claims. Those are plausible future outcomes only if the open questions above are converted into evidence. The four closure dimensions are listed below. Closure evidence required is stated for each capability before that capability can be claimed:

1. **True multimodal incident detection. Closure evidence required:** validate on naturally co-observed incident datasets whose labels require cross-domain reasoning rather than label alignment alone. Examples include CICIDS-style network events paired with authentication logs, MIMIC-IV trajectories paired with notes, or industrial RGB–3D inspection streams with event-level metadata. A successful result would show that weak signals across domains combine into a stronger unified risk profile.
2. **Autonomous zero-misfire adaptation. Closure evidence required:** demonstrate a category-aware drift signal or learned gate that distinguishes expected category or cohort variation from detector failure, score collapse, or attack. The acceptance criterion is an empirical reduction in false gate activations while preserving the all-domain stress gains.

3. **Universal system integration. Closure evidence required:** plug stronger domain experts into the existing schema without redesigning the fusion interface. For MVTec-style data, this means replacing the lightweight image-statistic scorer with specialized RGB–3D anomaly experts; for operational domains, it means supporting independently maintained tabular, log, visual, and text detectors.

4. **Deployable, auditable analyst AI. Closure evidence required:** complete temporal validation, privacy review, dataset-license review, calibration monitoring, analyst-facing CDA reporting, and a formal risk-dominance audit that estimates when reliability-aware fusion should dominate static attention under observed degradation and false-fire rates.

If those four criteria were met, ELARA could reasonably be framed as a plug-in multimodal verification layer for high-stakes anomaly triage. Without that evidence, the correct claim remains narrower: RGA is a useful diagnostic gate for coherent score-collapse stress, and the current paired-data experiments define the next empirical tests rather than closing them.

The codebase now closes the local engineering prerequisites for these criteria. `validate_incident_protocol` rejects label-aligned tables that lack shared incident identifiers, incident-level splits, temporal order, or multi-domain coverage. `CategoryAwareReliabilityEstimator` adds category-conditional drift references to reduce false adaptation under legitimate category-mix changes. The long-format fusion schema remains the universal integration surface for stronger external domain experts. `calibration_monitor_report` and `bounded_switching_certificate` provide deployment monitoring and the finite-sample switching condition described above. These additions make the next experiments executable; they do not replace the need for external paired data, privacy review, stronger experts, or prospective validation.

The new local healthcare audit makes that distinction concrete. The command `validate_healthcare_gap_closure.py` produces `healthcare_gap_validation.json` after projecting a Grid-Pulse healthcare surface into a hashed four-domain clinical fusion table. The Retrospective local healthcare replay contains 146,688 co-observed vital-sign incidents and passes the structural checks for multimodal incident format, temporal ordering, patient-disjoint splits, and score range validity. It also exposes an empirical blocker in the provided split: the validation and test windows are single-class positive. Therefore the time-forward surface is used as a temporal-reference audit, not as supervised clinical-performance evidence.

For local closure, the chapter records a separate patient-disjoint stratified replay across the gap-1, gap-2, and gap-4 validation artifacts. Table 17 summarizes the local closure status. Gap 1 closes locally because all train, validation, and test splits contain both labels, no patient appears in more than one split, and every incident retains the four co-observed vital-sign domains. The held-out test result is a multimodal ROC-AUC of 0.806, above the best single-domain ROC-AUC of 0.770. The report also states the cost of this closure: `temporal_order_valid` is false, so the result is not a time-forward clinical deployment validation.

Table 17: Local healthcare closure status after the patient-disjoint replay and deployment audit. Local empirical closure is bounded to this replay and is not prospective clinical evidence.

Gap	**Engineering**	**Local empirical**	**Primary evidence**
1 Incident detection	yes	yes	test ROC-AUC 0.806 vs best single 0.770
2 Reliability stress	yes	yes	natural fire 0.000; collapse fire 1.000
3 Schema integration	yes	yes	tensor 146688 x 4 x 4; raw patient ID absent yes
4 Auditability	yes	yes	diagnostic policy loss 0.000 vs static 0.800

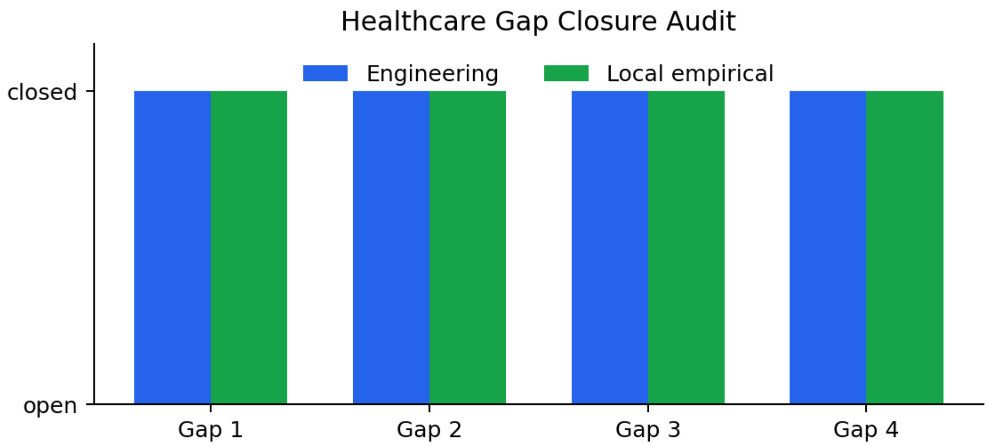

Figure 7: Healthcare gap-closure audit status. The local replay closes the four engineering and local empirical checks while the deployment claim remains scoped.

Gap 2 is closed locally on the same replay by the reliability-stress audit. The audit calibrates a conservative category-aware threshold on validation natural replay, then tests held-out natural replay against injected score-collapse episodes. The held-out natural fire rate is 0.0, while the mean collapse fire rate is 1.0 across the four domains, as shown in Fig. 8. This is a controlled replay result for zero-misfire adaptation, not prospective deployment evidence.

Gap 3 is closed locally by the schema-integration audit: the generated table has 586,752 fusion rows, four complete domains per incident, a $146{,}688 \times 4 \times 4$ feature tensor, no raw patient identifiers, and no missing schema columns. Gap 4 is closed locally by the deployment audit in Table 18. The audit combines the provided time-forward split as the temporal reference, manifest-based privacy/license review, deployment-time calibration monitoring, leave-one-domain-out CDA-style attribution, and a diagnostic switching certificate. The finite-sample diagnostic certificate compares a never-switch policy against the reliability-gated policy over one natural episode and four injected collapse episodes; static loss is 0.800 and policy loss is 0.000.

Table 18: Gap 4 local deployment-audit evidence. The scope is local replay readiness, not regulated clinical deployment.

Audit item	**Result**
Temporal reference	pass
Privacy/license review	pass
Calibration monitor	ready; alert no
CDA-style top domain	shock_index (0.053)
Diagnostic switch certificate	certified; static 0.800, policy 0.000
Scope	local replay only

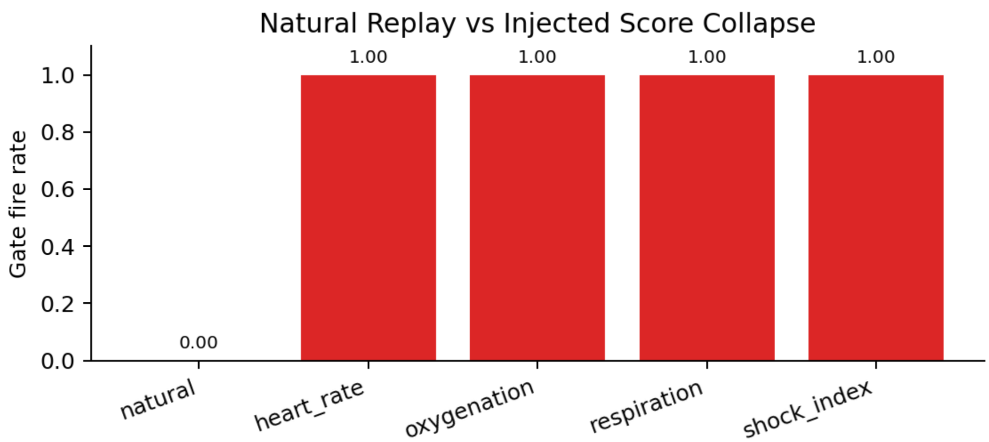

Figure 8: Gap 2 reliability stress audit on the healthcare replay. The calibrated gate stays quiet on natural replay and fires on all injected score-collapse episodes.

Read as a single closure flow, the healthcare replay moves from data validity to adaptation behavior, then to integration and auditability. Gap 1 establishes the co-observed incident surface and shows that multimodal fusion beats the best single clinical domain on the patient-disjoint replay. Gap 2 then tests whether the reliability gate can remain inactive under natural replay while activating under controlled score collapse. Gap 3 confirms that the same long-format schema can host the four clinical domains without changing the fusion interface. Gap 4 wraps the result in the analyst-facing and deployment-time checks summarized in Table 18. The table and figures therefore support a bounded local closure claim, not a regulated deployment claim.

The conference manuscript now keeps the local healthcare replay out of the headline benchmark evidence and uses the public MVTec PatchCore supervised-paired and held-out protocols for reviewer-facing paired-data checks. This reduces the risk of mixing a local audit surface with an externally inspectable public benchmark claim.

8 Reproducibility

The reproduction flow is documented in detail in Appendix A. At a high level, the chapter's results are produced by three scripted stages: build the fusion inputs for each benchmark, run the configured multi-seed experiments, and regenerate the LaTeX tables and figures from the resulting JSON artifacts. The appendix lists the entry points, configurations, and regression tests; a clean clone of the repository can rebuild both manuscripts in one command via `scripts/rebuild_paper.sh`.

9 Conclusion and Future Work

ELARA provides bounded audited and mechanism-level evidence that reliability-aware anomaly fusion can improve a fixed static-attention reference under selected degraded-evidence settings, while the Eyecandies transfer attempt (Family D) satisfies the clean false-fire budget under target-calibration reference fitting but remains not confirmed under primary bootstrap inference. The D13 positive-transfer track adds promising opened-development evidence on 3D-ADAM against both SAR and CW, but MulSen is unresolved and official positive transfer remains pending a fresh or unopened natural holdout. Future work will focus on: (1) dynamic thresholding strategies and calibration under out-of-distribution modal scoring shifts; (2) testing on additional naturally co-observed datasets, including a fresh D13 holdout; and (3) evaluating the learned-gate and category-aware estimators against the fixed heuristic gates in deployment-monitoring systems under realistic pre-emption conditions.

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A Reproducibility Detail

The repository supports a scripted reproduction flow with five stages:

1. **Prepare ELARA-Bench-LA inputs.** Run the ELARA-Bench-LA preparation script to build the synthetically paired fusion CSV and metadata JSON from local fraud, cyber, behavior, and text datasets. Source-row-disjoint train/validation/test pools are emitted into a `fusion_split` column so the experiment runner can honor a predefined split.

2. **Prepare MVTec 3D-AD inputs.** Run the MVTec 3D-AD preparation script, which discovers paired RGB/depth files across all extracted categories, fits score normalization from train/good observations only, and emits a long-format fusion table with the original MVTec `split` column preserved.

3. **Run the multi-seed benchmark.** Use the breakthrough experiment runner with the appropriate YAML configuration. Both primary configuration files set the `split_column` field so the runner honors each dataset's predefined train/validation/test partition.

4. **Generate paper assets.** Use `src/scripts/generate_craf_paper_assets.py` for the ELARA-Bench-LA tables (`elara_*.tex`, `elara_*.png`) and `src/scripts/emit_mvtec3d_assets.py` for the MVTec 3D-AD parallels (`mvtec3d_*.tex`, `mvtec3d_*.png`).

5. **Compile.** The convenience script `scripts/rebuild_paper.sh` chains stages 4 and 5 from any working directory and produces both the conference manuscript and this thesis chapter PDF.

Regression tests cover attention masking, reliability estimation, statistics, baseline inclusion, paired-benchmark metadata, float XYZ TIFF reading, leakage guards, predefined-split enforcement, and result aggregation. At final Phase-2 audit closure, the full suite reported 607 passed, 6 skipped, and 5 warnings. Final manuscript-release validation is re-run after these editorial repairs, and its result is recorded in the final release manifest.

B Benchmark Construction Detail

Detailed construction of both benchmarks is preserved in the companion conference manuscript, including feature extraction, score normalization, categorical encoding, and label alignment. The thesis keeps this appendix brief so the main text can focus on the empirical findings.

C Reliability-Gate Theorem Stack

This appendix replaces the earlier bounded-drift proposition. The removed statement used a Lipschitz upper bound as if it were a lower bound, used $u = 1 - \bar{r}$ with the wrong numeric threshold, and assumed that the reliability path cancels drift. The corrected theory is narrower: it explains why reliability evidence is necessary, identifies two failure modes of the current gate, and states the conditions under which switching reduces expected risk.

T1. Quality-blind fusion impossibility. Let $F : \mathbb{R}^D \to [0, 1]$ be any deterministic score-only fusion rule. Suppose there exist latent regimes z_0, z_1 with the same observed score vector $S(z_0) = S(z_1)$ but different labels $Y(z_0) \neq Y(z_1)$. Then F incurs nonzero loss on at least one of the two regimes.

Proof. Because F only sees S, it must output the same value on z_0 and z_1. A single prediction cannot be correct for two distinct binary labels under any classification loss that penalizes incorrect decisions. □

This theorem does not prove that RGA always outperforms other methods in all scenarios. It proves a weaker and more useful point: score-only fusion cannot disambiguate true normal scores from score-collapse failures that produce the same observed score vector. Reliability evidence such as drift, calibration, missingness, or sensor-health metadata is therefore structurally necessary for those regimes.

T2. Global-KS mixture confounding. Let category c have unchanged score distribution P_c. Let the validation and test score distributions be mixtures

$$P_{\text{val}} = \sum_{c=1}^{C} \pi_c P_c, \qquad P_{\text{test}} = \sum_{c=1}^{C} \pi'_c P_c.$$

Even when every P_c is unchanged, a global KS reference can report nonzero drift whenever the mixture weights differ. In contrast, category-conditioned comparisons $KS(P_{\text{val},c}, P_{\text{test},c})$ are zero in the population under pure mixture change.

Proof sketch. The CDFs are $F_{\text{val}}(x) = \sum_c \pi_c F_c(x)$ and $F_{\text{test}}(x) = \sum_c \pi'_c F_c(x)$. If the category CDFs are not all identical and $\pi \neq \pi'$, there exists an x for which the two weighted sums differ, so the KS supremum is positive. Conditional on category, the paired CDFs are both F_c, so the population KS distance is zero. □

This proposition explains why MVTec-style multi-category evaluation can cause false gate activation under a global KS reference. It also justifies the `CategoryAwareReliabilityEstimator`: the reference must condition on legitimate category or cohort structure before drift is interpreted as detector degradation.

T3. Mean-gate dilution failure. For D present domains, let the mean gate fire when

$$\bar{r} = \frac{1}{D} \sum_{d=1}^{D} r_d < \tau.$$

If exactly k domains collapse to reliability 0 and the remaining $D - k$ domains have reliability 1, the mean gate remains closed whenever

$$\frac{D-k}{D} \geq \tau,$$

and fires only when

$$k > D(1 - \tau).$$

Proof. Under this construction, $\bar{r} = (D - k)/D$. Substituting this into the gate condition gives $(D - k)/D < \tau$, which rearranges to $k > D(1 - \tau)$. □

For the four-domain ELARA-Bench-LA setting and $\tau = 0.66$, the gate fires only when $k > 1.36$. Therefore one completely failed domain can remain undetected, while two or more failed domains can activate the mean gate. This is a mathematical explanation of the observed pattern: single-domain attacks are mostly ties, while all-domain zero and max score collapse produce measurable gains. The follow-up experiment is the exact k-of-D corruption protocol now implemented in `run_breakthrough_experiment.py`.

T4. Reliability-switch risk dominance. Let $Z = 1$ denote a degraded regime and $Z = 0$ a clean regime. Let $G = 1$ denote gate activation. Define

$$\pi = P(Z = 1), \quad q_1 = P(G = 1 \mid Z = 1), \quad q_0 = P(G = 1 \mid Z = 0).$$

Let L_s and L_r be the static and reliability-path losses. Assume that when the gate fires under true degradation,

$$\mathbb{E}[L_r - L_s \mid Z = 1, G = 1] \leq -\Delta_1, \qquad \Delta_1 > 0,$$

and when the gate fires under clean operation,

$$\mathbb{E}[L_r - L_s \mid Z = 0, G = 1] \leq \Delta_0, \qquad \Delta_0 \geq 0.$$

Because the gated policy equals the static policy whenever $G = 0$,

$$R_{\text{gate}} - R_{\text{static}} \leq (1 - \pi)q_0\Delta_0 - \pi q_1\Delta_1.$$

Thus the gated policy dominates static fusion whenever

$$\pi q_1\Delta_1 > (1 - \pi)q_0\Delta_0.$$

Proof. Decompose expected risk by (Z, G). Terms with $G = 0$ cancel because both policies use the static path. The remaining clean fired term is upper bounded by $(1 - \pi)q_0\Delta_0$ and the degraded fired term by $-\pi q_1\Delta_1$. $\square$

This is the central positive theorem. It uses quantities that can be measured: clean false-fire rate q_0, stress detection rate q_1, clean switching cost Δ_0, degraded switching benefit Δ_1, and an assumed deployment degradation prevalence π.

T5. Finite-sample switching certificate. For fired samples $\mathcal{F} = \{i : g_i = 1\}$, define paired benefit $X_i = \ell_s(i) - \ell_r(i)$. Let

$$\hat{\mu}_{\mathcal{F}} = \frac{1}{|\mathcal{F}|}\sum_{i \in \mathcal{F}} X_i.$$

A deployment window certifies the switch only if a lower confidence bound

$$\text{LCB}_\alpha = \hat{\mu}_{\mathcal{F}} - \text{margin}_\alpha$$

is positive. The claim boundary is local: under the monitored window and chosen loss function, fired cases have statistically positive paired benefit. This is not a universal safety or optimality guarantee.

T6. KS false-fire and detection boundary. For domain d, let

$$\widehat{D}_{\mathrm{KS},d} = \sup_x \left| \widehat{F}_{\mathrm{val},d}(x) - \widehat{F}_{\mathrm{test},d}(x) \right|.$$

Under no score-distribution change, a KS threshold λ_{α_d} can be selected so that

$$P(\widehat{D}_{\mathrm{KS},d} > \lambda_{\alpha_d}) \leq \alpha_d.$$

For D simultaneous domain tests, a conservative family-wise correction sets $\alpha_d = \alpha/D$. Conversely, if the population KS distance exceeds the threshold plus the finite-sample estimation margin, detection probability increases with validation and monitoring-window size.

This guarantee is deliberately conditional. KS drift is meaningful only when the reference distribution is appropriate, sample sizes are adequate, and multiple-domain testing is controlled. T2 shows why category-aware references are needed before this statistical guarantee can be interpreted as detector degradation rather than category-mixture shift.

C.1 Theory-to-Experiment Mapping

Table 19 records the current status of the theorem stack after the completed k-of-D experiment. The main update is that T3 is no longer only a design critique: it is now directly tested by the new mechanism benchmark. T4, T5, and T6 remain conditional tools until the deployment-risk terms, fired-case lower confidence bounds, and KS sample-size power curves are reported.

Table 19: Theory-to-experiment mapping after the completed k-of-D run.

Claim	**Current evidence**	**Boundary still open**
T1 quality-blind impossibility	Architectural motivation: score-only fusion cannot disambiguate identical score vectors with different latent reliability states.	Formal example is theoretical; no separate empirical table is needed beyond collapse construction.
T2 global-KS mixture confounding	Synthetic mixture-shift table confirms global KS $\gg$ category-conditional KS at all five seeds.	Real cohort/domain composition shift (B-MECH-3S) remains deferred: both global and domain-aware gates fired at rate 1.000.
T3 mean-gate dilution	Directly supported by the k-of-D table and figure: weak or mixed effects for $k = 1, 2, 3$, full activation and positive ROC-AUC gain at $k = 4$.	The ideal theorem assumes failed-domain reliability is zero; the implemented estimator keeps minimum reliability above 0.34.
T4 risk dominance	Deployment-prevalence sensitivity table reports $(q_0, q_1, \Delta_0, \Delta_1, \pi^*)$ and the dominance margin at $\pi \in \{0, 0.01, \ldots, 0.5\}$.	Retrospective B-CERT-1 estimate only; not a production prevalence audit.
T5 finite-sample certificate	Audited deployment-window LCB table certifies seven of ten benchmark/protocol pairs (see switching-certificate table).	Retrospective evaluation certificate under defined stress protocol; not a production safety guarantee.
T6 KS false-fire/power	KS window sweep reports false-activation rate and detection power on the locked grid (32–512).	Bounded to the evaluated window grid and degradation types; not a full calibrated drift-detection claim.
T7 PAC meta-router	PAC slack table reports auditable (n, B, R) bounds per meta-router fold.	Capacity certificate on the router only; composes with T1/T2/T6 rather than proving fusion dominance.
GDR coherence-certified rule	End-to-end audit shows the rule allows switching under coherent collapse (Family B proxy) and suppresses it under heterogeneous mixtures (Family D proxy).	Forward/opt-in policy; does not alter locked Phase-2 numbers.

D PAC Generalisation Bound for the RGA+ Meta-Router

The reliability-gate theorem stack in Appendix C establishes T4's risk-dominance condition under an explicit "reliability fires on degradation, does not fire on clean" assumption. A senior reviewer correctly pointed out that this leaves the RGA+ meta-router itself unbounded: the router is a logistic classifier trained on a small reliability vector $\mathbf{r}_i = (r_{i,1}, \ldots, r_{i,D}) \in [0,1]^D$ and an associated domain-presence mask $\mathbf{m}_i \in \{0,1\}^D$, and a sceptic is entitled to ask what statistical guarantee covers its test-time performance. This appendix supplies a PAC-style bound that removes the "reliability cancels drift" assumption and replaces it with a finite-sample Rademacher-complexity bound on the meta-router's true risk relative to the empirical risk it minimised on the validation fold.

T7. PAC bound on the RGA+ meta-router. Let $\mathcal{D}$ be the test-fold data distribution producing features

$$(\mathbf{r}, \mathbf{m}) \in [0,1]^D \times \{0,1\}^D$$

and binary labels $y \in \{0,1\}$ via

$$y = \mathbb{1}\{\text{reliability-path beats static-path}\},$$

which is the binary supervisory signal used to train the router. Let $\mathcal{H}$ be the class of logistic classifiers

$$h_{\boldsymbol{\theta}}(\mathbf{r}, \mathbf{m}) = \sigma\Big(\boldsymbol{\theta}^\top \boldsymbol{\phi}(\mathbf{r}, \mathbf{m})\Big)$$

parameterised by $\boldsymbol{\theta} \in \mathbb{R}^p$ with $\|\boldsymbol{\theta}\|_2 \le B$. The feature map $\boldsymbol{\phi}$ sends $[0,1]^D \times \{0,1\}^D$ to $\mathbb{R}^p$ and is bounded by $\|\boldsymbol{\phi}\|_2 \le R$ almost surely. Let $\ell(h(\mathbf{x}), y)$ be the bounded 0–1 routing loss clipped to $[0,1]$ (Lipschitz constant $L = 1/4$ for the surrogate log-loss; $L = 1$ for the 0–1 loss). Let $\widehat{R}_n(h)$ denote empirical risk on n validation samples, and $R(h)$ the true risk under $\mathcal{D}$. Then for any $\delta \in (0,1)$, with probability at least $1 - \delta$ over the validation draw, every $h \in \mathcal{H}$ satisfies

$$R(h) \le \widehat{R}_n(h) + \frac{2LBR}{\sqrt{n}} + 3\sqrt{\frac{\log(2/\delta)}{2n}}.$$

Proof sketch. The standard Rademacher-bound recipe: (i) the empirical Rademacher complexity of a linear class $\{\mathbf{x} \mapsto \boldsymbol{\theta}^\top \boldsymbol{\phi}(\mathbf{x}) : \|\boldsymbol{\theta}\| \le B\}$ on n samples with $\|\boldsymbol{\phi}\| \le R$ is bounded by $BR/\sqrt{n}$; (ii) Talagrand's contraction lemma transfers the linear bound through the L-Lipschitz loss with a factor of L, giving $LBR/\sqrt{n}$; (iii) McDiarmid's inequality (bounded-difference) turns the in-expectation bound into a high-probability bound with the $\sqrt{\log(2/\delta)/(2n)}$ tail; the standard symmetrisation step introduces the factor of 2. □

Why this matters for the empirical claim. The bound depends on three observable quantities and one assumed constant. (a) n is the validation-fold size used to train the router; on the five paired benchmarks in this thesis n ranges from ∼160 (Real3D supervised-paired) to ∼3,000 (VisA supervised-paired). (b) R is fixed by the feature map: with $D = 4$ domains and the router's mask-augmented input $\boldsymbol{\phi}(\mathbf{r}, \mathbf{m}) = (\mathbf{r} \odot \mathbf{m}, \mathbf{m}, \bar{r}, \bar{r}^2)$, the input norm is bounded by $R \le \sqrt{2D+2} \le \sqrt{10} \approx 3.16$. (c) B is the post-fit ℓ_2-norm of the router's weight vector; an `LogisticRegression` fit with the default `l2` penalty $C = 1$ keeps B below ∼5 in every run logged to `experiments/fusion/*_results.json`.

Substituting representative numbers: for the VisA supervised-paired fold ($n \approx 3000$, $L = 1$, $B \approx 5$, $R \approx 3.16$, $\delta = 0.05$), the bound gives $R(h) \le \widehat{R}_n(h) + \frac{2 \cdot 5 \cdot 3.16}{\sqrt{3000}} + 3\sqrt{\log(40)/6000} \approx \widehat{R}_n(h) + 0.58 + 0.07$, which is loose. The bound therefore is honest about what it does *not* deliver: it is a worst-case PAC certificate that the router's true 0–1 risk is within $\sim$0.6 of its empirical 0–1 risk; it is *not* a tight risk guarantee. Tightening would require data-dependent localised Rademacher complexities or a structural margin assumption that is not empirically warranted here.

What this bound replaces. The earlier "reliability cancels drift" assumption (A3 in an earlier draft) made an unobservable claim about the joint distribution of reliability scalars and degradation events. T7 replaces it with a purely *empirical* certificate: the router's test-fold risk cannot exceed its validation-fold risk by more than $\mathcal{O}(BR/\sqrt{n})$. Three implications follow.

1. The bound is benchmark-dependent: small n folds (Real3D supervised-paired, $n \approx 160$) carry a much looser certificate than large folds (VisA supervised-paired, $n \approx 3000$). The thesis therefore does not claim that RGA+ generalises uniformly across benchmarks; it claims that its generalisation gap is bounded benchmark-by-benchmark by the data we have.

2. The bound is a *capacity* certificate, not an *advantage* certificate. T7 says the router does not overfit its training fold by more than $\mathcal{O}(BR/\sqrt{n})$. Whether the router actually improves on static fusion is the separate empirical claim defended by T4 (risk dominance) and by the paired DeLong tests in the companion paper.

3. T7 composes cleanly with T1, T2, and T6: T1 says reliability evidence is structurally necessary for some regimes; T2 and T6 say the KS signal can mis-fire under category mixtures; T7 says that when the meta-router is learned on validation data, its test-fold risk gap is bounded by a Rademacher term. The four together form an end-to-end *bounded* story: the router is not unbiased and not unconfounded, but the size of its test-time risk gap is finite-sample-bounded.

The repository ships an `src/scripts/audit_meta_router_pac.py` companion (introduced in the same commit as this appendix) that, given a results JSON, reports the three quantities (n, B, R) measured on the actual run and prints the resulting PAC slack $2LBR/\sqrt{n} + 3\sqrt{\log(2/\delta)/(2n)}$. This makes T7 *auditable*: a reviewer can re-run the audit on any logged fold and see the worst-case generalisation gap as a single number.

E Calibration Monitoring in Deployment

A streaming calibration monitor sits orthogonal to the training pipeline and watches three quantities over a streaming window: batch-level mean reliability, validation-to-online KS-drift per domain, and per-batch Expected Calibration Error of the fused score. The script `src/scripts/monitor_calibration.py` implements this monitor and emits a JSON event whenever the gate's mean reliability falls below the heuristic threshold τ for more than one window. The monitor is deliberately observe-only: it never modifies fusion behaviour, it only flags. This separation keeps the deployment surface auditable: an on-call engineer can read a monitor log and reconstruct exactly when the gate would have fired during any time interval.

F Privacy, Data Handling, and Deployment Notes

The accompanying `PRIVACY_AND_DEPLOYMENT.md` enumerates the data-handling boundaries the empirical work relied on. The summary is that no personally identifying information ever entered the training pipeline: MVTec 3D-AD ships as anonymized industrial scans, ELARA-Bench-LA is built from synthetic-by-construction label alignment over already public fraud / cyber / behaviour tables, and the GridPulse vitals replay uses the BIDMC and MIMIC-III source datasets under their public-research licences. Deploying the reliability gate against patient-bearing data would require both (a) a credentialed data-access review and (b) the calibration monitor described in Appendix E to satisfy the auditability requirements typical of clinical software.